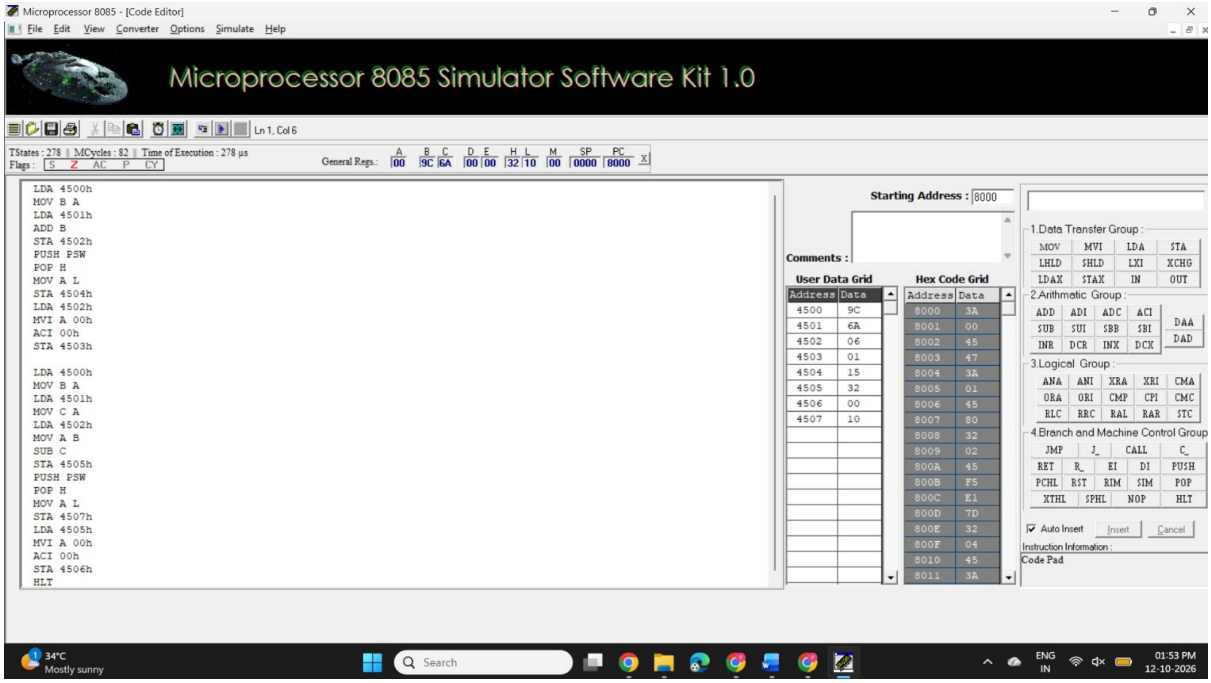
Name: Y.Varun Kumar Reddy

Reg no:192424230

EXPERIMENT – 1

1.

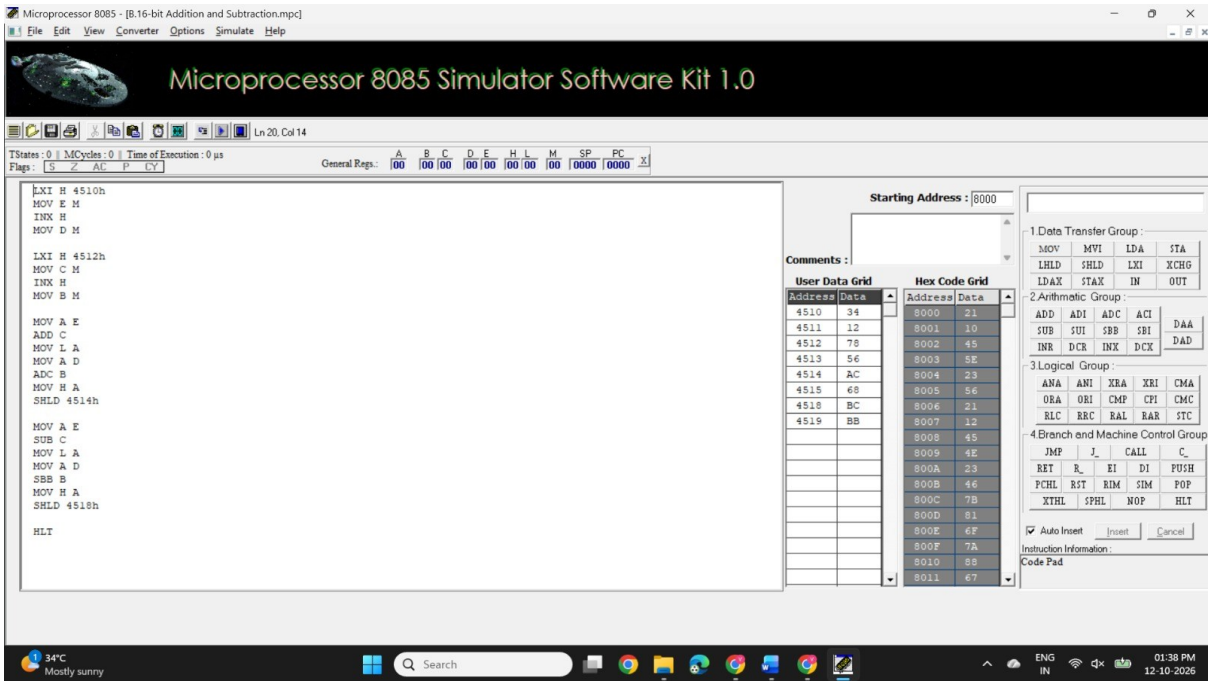
AIM: 8-bit Addition and Subtraction with Flag Analysis: Store A = 9CH at 4500H and B = 7AH at 4501H. Execute Program A from 4100H. The sum and carry are stored at 4502H–4503H and the difference and borrow at 4505H–4506H. PUSH PSW saves the accumulator and the flag register after each operation, and the flag byte (S Z X AC X P X CY) is stored at 4504H and 4507H. Predict every flag manually and compare.



2.

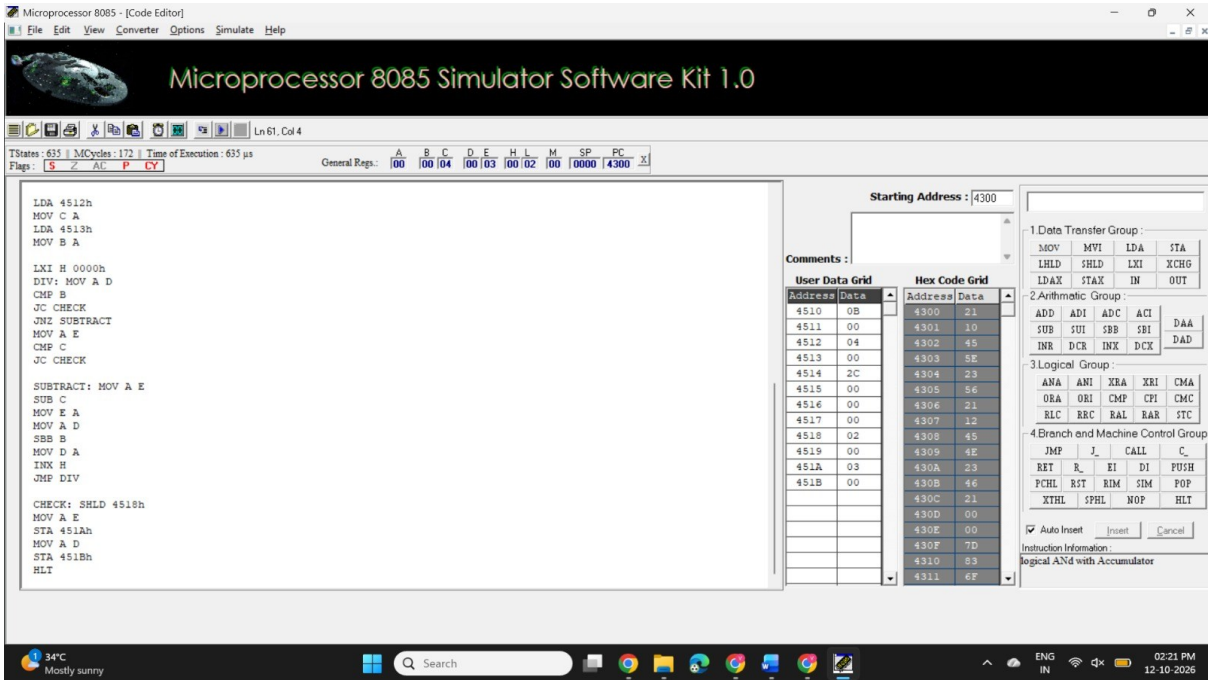
AIM: 16-bit Addition and Subtraction: Store X = 1234H at 4510H and Y = 5678H at 4512H (lower byte first). Execute Program B from 4200H. DAD D forms the sum, which is stored with the carry at 4514H–4516H; SUB and SBB form the difference, which is stored with the borrow at 4518H–451AH. Observe that X < Y produces a borrow and a two's-

complement difference



3.

8-bit Multiplication and Division: Store 12H at 4520H and 05H at 4521H; store the dividend 64H at 4524H and the divisor 05H at 4525H. Execute Program C from 4300H. The product is formed by repeated addition and stored at 4522H–4523H; the quotient and remainder are formed by repeated subtraction and stored at 4526H and 4527H.



4.

16-bit Multiplication and Division: Store X = 1234H at 4530H, Y = 5678H at 4532H, the dividend 5678H at 4538H and the divisor 1234H at 453AH. Execute Program D from 4400H. The 32-bit product is stored at 4534H (lower word) and 4536H (upper word); the quotient and remainder are stored at 453CH and 453EH. Verify N = Q × divisor + R.

